Lane Brief 1 — Japan and Germany Labels

Owner: Shravani Math **Role:** Data and label lead **Lane:** Part 1 of the one-week sprint **Dates:** Days 1–5 individual; Days 6–7 shared merge and review **Branch:** `feat/jpn-deu-labels`

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1. Your mission in one paragraph

Build the recession answer key for Japan with the **frozen** estimator, and resolve Germany far enough to either build its label or write a complete 6-of-8 coverage finding. Japan is the priority. Germany is a stretch goal, and a complete coverage note is an acceptable deliverable. You also own the small refactor that lets the existing USA pipeline run for another country without forking the label math.

2. What has already happened (so you do not repeat it)

- **The USA label is built, validated and frozen.** `labels-v3` covers 1990Q1–2026Q3 (147 quarters), matches all four NBER recessions since 1990, and has zero false episodes. Accuracy is 0.993, precision 0.929, recall 1.000, F1 0.963.
- **The estimator is frozen.** One-factor `DynamicFactorMQ`, smoothed factor, BBQ with min phase 2, min cycle 5, endpoint censoring 2, diffusion gate 0.50. D-R22–D-R31 record the decisions. Do not change them for Japan or Germany.
- **The USA data choices are settled.** Q-02 uses constructed real non-residential business fixed investment; Q-04 uses `CE160V`; Q-05 keeps `HOANBS` with its coverage limitation documented. These are USA choices, not automatic choices for Japan or Germany.
- **`labels-v1` and `labels-v2` are archived byte-identically.** `labels-v3` is active. The freeze guard refuses to overwrite a frozen label when the canonical input digest changes.
- **The country plumbing is not general yet.** `scripts/harmonize.py` and `scripts/03_labels.py` currently hard-code the USA. The label builder itself (`src/ews/labels/build.py`) accepts a `country` argument, so the estimator is reusable. Generalising the scripts — or adding thin JPN/DEU wrappers — is your Day 1 task.
- **Japan and Germany data are not collected yet.** The scope has 21 JPN and 21 DEU rows. The source IDs for JPN/DEU still need verification; do not guess them.
- **India is not your lane.** The India pre-audit finds a best case of 4/8, so India is a Label-B case study under O-2. Do not spend your week on India.

Read first: `docs/BUILD_CONTRACT.md` (C-01–C-13), `docs/DECISIONS.md` (D-R22–D-R31), `docs/LABEL_VALIDATION.md`, `scripts/03_labels.py`, `scripts/harmonize.py`, `src/ews/labels/build.py`, `src/ews/labels/freeze.py`.

3. What you own: Days 1–5

Day	Task	Output	Done when
1	Map Japan sources **and** generalise the country plumbing	Japan source map; country-parameterised `scripts/harmonize.py` and `scripts/03_labels.py` (or wrappers); naming convention	21/21 Japan rows have a source or a written pending reason; USA `labels-v3` hash unchanged after the refactor
2	Download and harmonise Japan	Raw downloads, manifest rows, JPN levels/growth, coverage table	6/8 core indicators at ≥70% of 147 quarters (≥103) or a documented failure; no interpolation
3	Build and freeze the Japan label	`outputs/labels/JPN/*`, `docs/LABEL_VALIDATION_JPN.md`	Label reproducible; ESRI chronology comparison written; `labels-v1-JPN` frozen with hashes
4	Map and download Germany	DEU source map and clean panel	21/21 rows resolved or pending with a reason; 1990–91 reunification break documented
5	Germany coverage and label decision	DEU coverage note, label if eligible, hand-off note	Gate decision recorded; no rule relaxation; hashes listed

Suggested daily rhythm

- **09:30 — 15-minute stand-up.** Say what you finished, what you will do, and what is blocking you. Ask for help early.
- **Work in small commits.** One logical change per commit; run the relevant tests.
- **17:30 — short hand-off note.** What changed, what is reproducible, what is still open.
- **Day 6:** merge your own branch in the fixed order. **Day 7:** cross-review and sign off.

4. What we expect by Day 5

A complete Part 1 hand-off contains:

1. **Japan source map.** One active source row per scope key, with frequency, seasonal adjustment, first/last date, retrieval date and SHA-256.
2. **Japan clean panel.** Quarterly, complete quarters only, no interpolation, no forward fill. The coverage table shows the 6/8 count and the window share.
3. **Japan label.** `outputs/labels/JPN/JPN_labels.csv`, episodes, turning points and a metadata JSON. The estimator, BBQ and diffusion settings are the frozen ones.
4. **Japan chronology note.** `docs/LABEL_VALIDATION_JPN.md`: method, ESRI comparison, discrepancies, caveats. Say "validated label", not "forecast".
5. **Germany decision.** Either a frozen German label if 6/8 passes, or a complete coverage note naming the missing indicators and the evidence. Do not fake a label.
6. **Evidence that the USA did not move.** Re-run the USA label hash check after the country-plumbing refactor. `labels-v3` must stay byte-identical.

7. **A PR on `feat/jpn-deu-labels`** with the commands run, tests passed, hashes changed and known gaps.

How your lane is judged

- **Label gate:** Japan (and Germany if attempted) needs ≥6 of 8 core indicators at ≥70% of the 147-quarter window (≥103 quarters), plus a sensible chronology check.
- **Reproducibility:** a fresh run from the raw files and the documented commands produces the same label and hashes.
- **Honesty:** a failed gate is written down clearly. A missing indicator is a finding, not an exception.
- **No collateral change:** the USA label, the estimator and the eight-indicator structure are unchanged.

5. How your work combines with the others

Hand-off	When	What must be true
H1: Sathvik → all	Day 1 morning	You can state your Day 1 task from this brief and `docs/TEAM_PLAN.md`
H3: you → Sathvik	Day 5 end	Japan label + chronology note + Germany 6/8 decision; hashes recorded
H6: all → Sathvik + Arshiya	Day 6 morning	One branch, green checks, short hand-off note
Day 6 merge	Step 1	You merge `feat/jpn-deu-labels` first; the team checks label hashes and chronology
Day 7 review	—	Arshiya cross-reviews your label lane; write findings, not verbal comments

Your outputs feed Part 2 (clean activity panels), Part 3 (labels as the target) and Part 4 (the final story). If Japan lands but Germany does not, that is still a usable result: Japan becomes the second country, Germany becomes a documented coverage finding.

6. DOs

- **DO** read `docs/BUILD_CONTRACT.md` and `docs/DECISIONS.md` before writing code.
- **DO** generalise the pipeline by parameter, not by copying the estimator math.
- **DO** keep the USA label byte-identical and prove it with a hash check.
- **DO** use official sources (BOJ, e-Stat, Cabinet Office, OECD, Destatis, BIS) and record the exact series ID, retrieval date and SHA-256.
- **DO** apply the complete-quarter rule and leave missing values missing.
- **DO** write a short note for every non-obvious choice and every failed source.
- **DO** add or update tests for the country plumbing and the coverage gate.
- **DO** ask for help on Day 1 if the refactor or a source key is unclear.
- **DO** commit small and keep the branch rebased on `main`.

7. DON'Ts

- **DON'T** change the estimator, BBQ parameters, diffusion gate or the eight-indicator structure without a new version and a recorded decision.

- **DON'T** relax the 6-of-8 rule or hide a failed country.
- **DON'T** interpolate, forward-fill or impute a partial quarter.
- **DON'T** overwrite `labels-v3`, `labels-v2` or `labels-v1`.
- **DON'T** fork the label math into a separate JPN script that will drift.
- **DON'T** commit restricted raw data; commit metadata, manifests and hashes only.
- **DON'T** claim a forecast or a cross-country result before the out-of-sample work exists.
- **DON'T** silently take over another lane or merge a red branch.

8. Allowed creative liberty and educational freedom

You are not a robot following a checklist. Inside the guardrails:

- **Choose your implementation.** Parameterising the existing scripts and adding thin wrappers are both acceptable. Pick the one you can test and explain.
- **Add useful extras.** A source-map table, a coverage plot, a short "Japan data notes" page, a glossary of Japanese series names, or a small notebook are welcome.
- **Learn while you work.** You may spend up to about half a day (roughly 10% of the lane) on the DFM, `statsmodels`, Japanese source documentation or a reproducibility tool. Note it in your hand-off.
- **Try an alternative source** if the first choice is blocked, as long as you document the substitution and do not relax the coverage rule.
- **Propose a better interface** for country labels before you build it. If it changes a shared output, raise it at the stand-up and record it.
- **Write a "what I learned" section** in your chronology note. It helps the next person.

9. If you get stuck

1. **Source key unclear?** Write the exact question and the candidate IDs in your note; ask the team before guessing.
2. **Refactor getting large?** Protect Japan first. A thin wrapper that reuses the estimator is better than a perfect abstraction that misses the week.
3. **Coverage fails?** Write the missing indicators, the window share and the evidence. That is a real result, not a failure of the lane.
4. **USA hash moves?** Stop. Fix the cause before continuing. Never patch the hash or the label by hand.
5. **Blocked for more than half a day?** Say so at the stand-up. Do not hide it until Day 5.

10. Where to look

Need	File
Frozen rules	`docs/BUILD_CONTRACT.md`
Decisions and rationale	`docs/DECISIONS.md`
USA label evidence	`docs/LABEL_VALIDATION.md`
Q-02/Q-04/Q-05 detail	`docs/DECISION_NOTE_Q02_Q04_Q05.md`
USA label builder	`scripts/03_labels.py`, `src/ews/labels/build.py`

Need	File
Freeze guard	`src/ews/labels/freeze.py`
Harmoniser	`scripts/harmonize.py`
Scope and sources	`config/scope.yaml`, `data/scope.csv`, `data/series_resolution.csv`
Team plan	`docs/TEAM_PLAN.md`, `Team_Plan.xlsx`
Weekly log	`EWS_Work_Log.xlsx`

Bottom line: Japan is the deliverable; Germany is a bonus or a well-documented coverage finding. Reuse the frozen estimator, keep the USA untouched, and write down every source and every failure. A clear "we could not get 6/8, here is why" is a successful week.